Surds

Surds are just expressions with a "n^{th} root" sign.

Properties of Surds: (For $a, b \geq 0$)

- $\sqrt{ab} = \sqrt{a}\sqrt{b}$
- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$
- $(\sqrt[n]{a})^n = a$
- $(\sqrt{a})^2 = a$ where $a \geq 0$
- $b\sqrt{a} = \sqrt{b^2}\sqrt{a} = \sqrt{b^2 \times a}$

Simplification of Surds:

Surds with bigger numbers can be simplified by extracting out a *perfect square*.

Example:

- $\sqrt{147} = \sqrt{49 \times 3} = \sqrt{49} \times \sqrt{3} = 7\sqrt{3}$
- $\sqrt{32} = \sqrt{16 \times 2} = \sqrt{16} \times \sqrt{2} = 4\sqrt{2}$

Rationalization of Denominator:

When the denominator contains ***just*** square roots, we could get rid of the square roots by multiplying the top and bottom by surds:

$$\frac{1-\sqrt{3}}{\sqrt{2}} = \frac{1-\sqrt{3}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}(1-\sqrt{3})}{\sqrt{2}\times\sqrt{2}} = \frac{\sqrt{2}-\sqrt{6}}{2}$$

Multiply by 1 **(does not change the value of an expression)**

When the denominator contains ***multiple surds/constant and surd terms***, we multiply the top and bottom by the **conjugate of the denominator:**

Conjugate of "$1 - \sqrt{3}$" = $1 + \sqrt{3}$

Conjugate of "$\sqrt{5} - \sqrt{3}$" = $\sqrt{5} + \sqrt{3}$

Conjugate of "$2\sqrt{5} + 3\sqrt{3}$" = $2\sqrt{5} - 3\sqrt{3}$"

$$\frac{2+\sqrt{3}}{2\sqrt{3}-3\sqrt{2}} = \frac{2+\sqrt{3}}{2\sqrt{3}-3\sqrt{2}} \times \frac{2\sqrt{3}+3\sqrt{2}}{2\sqrt{3}+3\sqrt{2}}$$

Multiply by Conjugate of "$2\sqrt{3} - 3\sqrt{2}$"

$$= \frac{2 + \sqrt{3}}{2\sqrt{3} - 3\sqrt{2}} \times \frac{2\sqrt{3} + 3\sqrt{2}}{2\sqrt{3} + 3\sqrt{2}}$$

$$= \frac{(2 + \sqrt{3})(2\sqrt{3} + 3\sqrt{2})}{(2\sqrt{3} + 3\sqrt{2})(2\sqrt{3} - 3\sqrt{2})}$$

$$= \frac{4\sqrt{3} + 6\sqrt{2} + 6 + 3\sqrt{6}}{-6}$$

Multiplication of surd conjugates is done using the algebra formula: $(a + b)(a - b) = a^2 - b^2$:

$$(2\sqrt{3} + 3\sqrt{2})(2\sqrt{3} - 3\sqrt{2}) = (2\sqrt{3})^2 - (3\sqrt{2})^2$$
$$= (4)(3) - (9)(2)$$
$$= -6$$

Rationalization of More Complicated Denominator:

➢ $\frac{2}{(3-\sqrt{2})^2} = \frac{2}{3^2-(2)(3)(\sqrt{2})+(\sqrt{2})^2}$

Expand out the denominator

$= \frac{2}{11-6\sqrt{2}} \times \frac{11+6\sqrt{2}}{11+6\sqrt{2}}$

Multiply the top & bottom by the conjugate

$= \frac{2}{11-6\sqrt{2}} \times \frac{11+6\sqrt{2}}{11+6\sqrt{2}}$

We can multiply the top and bottom by $(3+\sqrt{2})^2$

$= \frac{2(11+6\sqrt{2})}{11^2-(6\sqrt{2})^2} = \frac{22+12\sqrt{2}}{121-(6^2)(\sqrt{2})^2} = \frac{22+12\sqrt{2}}{49}$

Alternatively:

$$\frac{2}{(3-\sqrt{2})^2} = \frac{2}{(3-\sqrt{2})^2} \times \frac{(3+\sqrt{2})^2}{(3+\sqrt{2})^2} = \frac{2(3+\sqrt{2})^2}{(3-\sqrt{2})^2(3+\sqrt{2})^2} = \frac{2\left[3^2+2(3)(\sqrt{2})+(\sqrt{2})^2\right]}{\left[(3-\sqrt{2})(3+\sqrt{2})\right]^2}$$

Multiply by the conjugate of $(3-\sqrt{2})^2$: Which is "$(3+\sqrt{2})^2$"

$(3-\sqrt{2})^2(3+\sqrt{2})^2 = \left[(3-\sqrt{2})(3+\sqrt{2})\right]^2$

Using the law of indices

$= \frac{2(11+6\sqrt{2})}{\left[3^2-(\sqrt{2})^2\right]^2} = \frac{22+12\sqrt{2}}{49}$

➢ $\frac{\sqrt{3}}{1+\sqrt{2}-\sqrt{3}}$ ← Conjugate of "$(1+\sqrt{2})-\sqrt{3}$" is: $(1+\sqrt{2})+\sqrt{3}$

Multiply top & bottom by the conjugate of the denominator gives:

$$\frac{\sqrt{3}}{1+\sqrt{2}-\sqrt{3}} = \frac{\sqrt{3}}{1+\sqrt{2}-\sqrt{3}} \times \frac{1+\sqrt{2}+\sqrt{3}}{1+\sqrt{2}+\sqrt{3}} = \frac{\sqrt{3}(1+\sqrt{2}+\sqrt{3})}{(1+\sqrt{2}-\sqrt{3})(1+\sqrt{2}+\sqrt{3})}$$

Use $(a+b)(a-b) = a^2 - b^2$

$$= \frac{\sqrt{3}+\sqrt{2}\sqrt{3}+(\sqrt{3})^2}{(1+\sqrt{2})^2-(\sqrt{3})^2} = \frac{\sqrt{3}+\sqrt{6}+3}{(1+2\sqrt{2}+2)-3}$$

$$= \frac{\sqrt{3}+\sqrt{6}+3}{2\sqrt{2}} = \frac{\sqrt{3}+\sqrt{6}+3}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

Multiply by $\sqrt{2}$ (NO need to multiply by $2\sqrt{2}$)

Rationalize the denominator

$$= \frac{\sqrt{2}(\sqrt{3}+\sqrt{6}+3)}{2(2)} = \frac{\sqrt{6}+\sqrt{12}+3\sqrt{2}}{4}$$

$$= \frac{\sqrt{6}+2\sqrt{3}+3\sqrt{2}}{4}$$

Example 1: Find the value of k when: $\frac{2}{\sqrt{2}}\left(\frac{3\sqrt{54}}{4}-\frac{15}{\sqrt{150}}-\frac{14}{\sqrt{294}}\right)=k\sqrt{3}$

Solution:

$$\frac{2}{\sqrt{2}}\left(\frac{3\sqrt{54}}{4}-\frac{15}{\sqrt{150}}-\frac{14}{\sqrt{294}}\right)=k\sqrt{3}$$

Simplify the surds $\sqrt{54}, \sqrt{150}, \sqrt{294}$

$$\frac{2}{\sqrt{2}}\left(\frac{3\sqrt{9\times6}}{4}-\frac{15}{\sqrt{25\times6}}-\frac{14}{\sqrt{49\times6}}\right)=k\sqrt{3}$$

$$\frac{2}{\sqrt{2}}\left(\frac{9\sqrt{6}}{4}-\frac{15}{5\sqrt{6}}-\frac{14}{7\sqrt{6}}\right)=k\sqrt{3}$$

Rationalize the denominators

$$\frac{2}{\sqrt{2}}\left(\frac{9\sqrt{6}}{4}-\frac{15\sqrt{6}}{5(6)}-\frac{14\sqrt{6}}{7(6)}\right)=k\sqrt{3}$$

$$\frac{2}{\sqrt{2}}\left(\frac{9\sqrt{6}}{4}-\frac{3\sqrt{6}}{6}-\frac{2\sqrt{6}}{6}\right)=k\sqrt{3}$$

Simplify the like terms within the brackets

$$\frac{2}{\sqrt{2}}\left(\frac{17\sqrt{6}}{12}\right)=k\sqrt{3}$$

$$\frac{2}{\cancel{\sqrt{2}}}\left(\frac{17\sqrt{3}\cancel{\sqrt{2}}}{6}\right)=k\sqrt{3}$$

$$\frac{17\sqrt{3}}{3}=k\sqrt{3}\Longrightarrow k=\frac{17}{3}$$

Example 2:

Express $\left(\frac{2}{\sqrt{3}+2}\right)^2-\frac{2}{3-\sqrt{3}}+\left(1-\sqrt{3}\right)^2$ in the form of $a+b\sqrt{3}$.

Solution:

$$\left(\frac{2}{\sqrt{3}+2}\right)^2-\frac{2}{3-\sqrt{3}}+\left(1-\sqrt{3}\right)^2$$

Simplify the Surd form

$$=\frac{4}{\left(\sqrt{3}+2\right)^2}-\frac{2}{3-\sqrt{3}}\times\frac{3+\sqrt{3}}{3+\sqrt{3}}+\left(1-\sqrt{3}\right)^2$$

$$=\frac{4\left(\sqrt{3}-2\right)^2}{\left(\sqrt{3}+2\right)^2\left(\sqrt{3}-2\right)^2}-\frac{2\left(3+\sqrt{3}\right)}{9-3}+\left(1-2\sqrt{3}+3\right)$$

$$= \frac{4(3-2\sqrt{3}+4)}{[(\sqrt{3}+2)(\sqrt{3}-2)]^2} - \frac{3+\sqrt{3}}{3} + (1 - 2\sqrt{3} + 3)$$

$$= \frac{4(7-2\sqrt{3})}{(3-4)^2} - \left(1 + \frac{\sqrt{3}}{3}\right) + 4 - 2\sqrt{3} = 28 - 8\sqrt{3} - 1 - \frac{\sqrt{3}}{3} + 4 - 2\sqrt{3}$$

$$= 31 - 10\sqrt{3} - \frac{\sqrt{3}}{3}$$

$$= 31 - \frac{31\sqrt{3}}{3} \implies a = 31, b = -\frac{31}{3}$$

Example 3:

By first expressing $\frac{2+\sqrt{3}}{2-\sqrt{3}}$ in the form of $a + \sqrt{b}$. Find the exact value of $\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right)^2 + \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2$

Solution:

$$\frac{2+\sqrt{3}}{2-\sqrt{3}} = \frac{2+\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{(2+\sqrt{3})^2}{4-3} = \frac{7+4\sqrt{3}}{1} = 7 + 4\sqrt{3}$$

$$\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right)^2 + \left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)^2 = (7 + 4\sqrt{3})^2 + \frac{1}{(7+4\sqrt{3})^2}$$

$$= (49 + 56\sqrt{3} + 48) + \frac{1}{49 + 56\sqrt{3} + 48}$$

$$= (97 + 56\sqrt{3}) + \frac{1}{97 + 56\sqrt{3}}$$

$$= (97 + 56\sqrt{3}) + \frac{1}{97 + 56\sqrt{3}} \times \frac{97 + 56\sqrt{3}}{97 + 56\sqrt{3}}$$

$$= (97 + 56\sqrt{3}) + \frac{97 + 56\sqrt{3}}{97^2 - 56^2(3)}$$

$$= (73 + 40\sqrt{3}) + \frac{73 - 40\sqrt{3}}{1}$$

$$= (73 + 40\sqrt{3}) + (73 - 40\sqrt{3})$$

$$= 146$$

If any one of a, b, d and e are NOT rational numbers, then the "equality of surds" cannot be applied!

Equality of Surds:

If *rational numbers* a, b, d, e are such that:

$$a + b\sqrt{c} = d + e\sqrt{c}, \text{ then: } \quad a = d \text{ and } \quad b = e$$

This rule is used in finding unknown constants by comparison of non-surd and surd form respectively on both sides.

Example 1:

Given that $(2 + \sqrt{3})(3 - b\sqrt{3}) = a + 7\sqrt{3}$, find the value of a and b.

Solution:

Collate all the non-surds and surd terms

$$L.H.S = (2 + \sqrt{3})(3 - b\sqrt{3}) = 6 - 2b\sqrt{3} + 3\sqrt{3} - b(\sqrt{3})(\sqrt{3})$$

$$= \underbrace{(6 - 3b)}_{a} + \underbrace{\sqrt{3}(3 - 2b)}_{7\sqrt{3}}$$

By comparison (equality of surds) with $R.H.S$:

$$6 - 3b = a \quad \text{--- (1)}$$
$$3 - 2b = 7 \text{ --- (2)}$$

Solving for b (from equation (2)) gives: $b = -2$

Substitute $b = -2$ into equation (1) $\Rightarrow a = 6 - 3(-2) = 12$

Example 2:

Find the value of p and q for which: $\frac{2+\sqrt{10}}{2\sqrt{5}+3\sqrt{2}} = p\sqrt{2} + q\sqrt{5}$

When there is no unknowns for the fraction term, FASTEST way is to rationalize the denominator

Solution:

We will rationalize the denominator on the $L.H.S$:

$$L.H.S = \frac{2+\sqrt{10}}{2\sqrt{5}+3\sqrt{2}} = \frac{2+\sqrt{10}}{2\sqrt{5}+3\sqrt{2}} \times \frac{2\sqrt{5}-3\sqrt{2}}{2\sqrt{5}-3\sqrt{2}} = \frac{(2+\sqrt{10})(2\sqrt{5}-3\sqrt{2})}{(2\sqrt{5})^2-(3\sqrt{2})^2}$$

$$= \frac{4\sqrt{5} - 6\sqrt{20} + 2\sqrt{50} - 3\sqrt{20}}{20 - 18} = \frac{4\sqrt{5} - 6(\sqrt{4})(\sqrt{5}) + 2(\sqrt{25})(\sqrt{2}) - 3(\sqrt{4})(\sqrt{5})}{2}$$

Simplify the surds ($\sqrt{20}$ and $\sqrt{50}$)

$$= \frac{4\sqrt{5}-12\sqrt{5}+10\sqrt{2}-6\sqrt{5}}{2} = \frac{8\sqrt{5}+10\sqrt{2}}{2} = \underbrace{4\sqrt{5} + 5\sqrt{2}}_{q\sqrt{5} + p\sqrt{2}}$$

Comparing with $R.H.S \implies q = 4$ and $p = 5$

Example 3:

If $\frac{6+4\sqrt{3}}{a+b\sqrt{3}} = 1 - \sqrt{3}$, where a and b are rational numbers, find the value of a and b.

When there is unknown at the denominator, multiple the denominator to the other side.

Solution:

$$\frac{6+4\sqrt{3}}{a+b\sqrt{3}} = 1 - \sqrt{3}$$

$$6 + 4\sqrt{3} = (1 - \sqrt{3})(a + b\sqrt{3})$$

Multiply the denominator over

Expand out the $R.H.S$:

$$R.H.S = (1 - \sqrt{3})(a + b\sqrt{3}) = \text{a} + \text{b}\sqrt{3} - \sqrt{3}a - b(\sqrt{3})(\sqrt{3})$$

$$= \underbrace{(a - 3b)}_{6} + \underbrace{\sqrt{3}(b - a)}_{4\sqrt{3}}$$

Thus, by comparison:

$a - 3b = 6$ --- (1)

$b - a = 4$ ---(2)

Solving simultaneously gives: $a = -9, b = -5$

Example 4:

Square both sides to get rid of the square root containing the unknown

If $\sqrt{p + q\sqrt{7}} = \frac{9}{4-\sqrt{7}}$, where p and q are rational numbers, find the values of p and q.

Solution:

$$\sqrt{p + q\sqrt{7}} = \frac{9}{4-\sqrt{7}} \implies p + q\sqrt{7} = \left(\frac{9}{4-\sqrt{7}}\right)^2$$

Rationalize the denominator since the denominator has no unknown

$$p + q\sqrt{7} = \frac{81}{(4-\sqrt{7})^2} \times \frac{(4+\sqrt{7})^2}{(4+\sqrt{7})^2}$$

$$p + q\sqrt{7} = \frac{81}{(4-\sqrt{7})^2} \times \frac{(4+\sqrt{7})^2}{(4+\sqrt{7})^2}$$

$$p + q\sqrt{7} = \frac{81(4+\sqrt{7})^2}{[16-7]^2} = \frac{81(16+8\sqrt{7}+7)}{81} = 23 + 8\sqrt{7}$$

By comparison, $p = 23$ and $q = 8$

Finding unknown in an Equation involving surds:

Squaring an equation in the process of solving for an unknown sometimes introduces "unwanted" roots (a.k.a "extraneous roots") $\Longrightarrow$ **must check the solution after solving.**

Knowing when to square both sides in a "surd equation" is important
(Squaring at the wrong time makes the question unnecessarily complicated)

Consider the "simple" equation:

$x - 1 = 2 \ \ (x = 3)$

If we square both sides of the equation:

$$(x-1)^2 = 4$$

Now we take the square roots on both sides again:

$x - 1 = \pm 2 \Longrightarrow x = 3$ or $x = -1$

$x = -1$ is an "extraneous" root

Example 1:

Solve the equation: $\sqrt{3x+4} - 4 = 1$

Isolate the numbers and the surd term and the constants when <u>there is only 1 surd</u>

Solution:

$\sqrt{3x+4} - 4 = 1$

Isolate the surd on the $L.H.S$

$\sqrt{3x+4} = 5$

Square both sides to get rid of the square root

$3x + 4 = 25$

$3x = 21 \ \Longrightarrow x = 7$ (No extraneous roots)

Example 2:

Solve the equation: $2\sqrt{x-1} - x + 1 = \frac{x}{2}$

Solution:

$2\sqrt{x-1} - x + 1 = \frac{x}{2}$

Isolate the surd on the $L.H.S$

$2\sqrt{x-1} = \frac{3x}{2} - 1$

$$\left[2\sqrt{x-1}\right]^2 = \left(\frac{3x}{2} - 1\right)^2 \Longrightarrow 4(x-1) = \frac{9x^2}{4} - 3x + 1$$

$$16(x-1) = 9x^2 - 12x + 4 \Longrightarrow 9x^2 - 28x + 20 = 0$$

$$(9x - 10)(x - 2) = 0$$

$\therefore x = \frac{9}{10}$ or $x = 2$

$x = \frac{9}{10}$ is not a solution since $\sqrt{x-1}$ will not be defined (we cant square root a negative number)

Substitute $x = 2$ into the original equation:

$L.H.S = 2\sqrt{2-1} - 2 + 1 = 1$

$R.H.S = \frac{2}{2} = 1$

} $x = 2$ is a solution

Example 3:

Solve the equation: $1 = \sqrt{x+2} - \sqrt{3-x}.$ ← Bring one of the surd to the side of the constant before squaring

Solution:

$1 = \sqrt{x+2} - \sqrt{3-x}$

$\sqrt{x+2} = 1 + \sqrt{3-x}$ ↘ Bring the Surd form to the other side

$(\sqrt{x+2})^2 = \left[1 + \sqrt{3-x}\right]^2$ ↘ Square both sides

$x + 2 = 1 + 2\sqrt{3-x} + 3 - x$

$2x - 2 = 2\sqrt{3-x}$

$(2x-2)^2 = 4(3-x)$ ↘ Square both sides again to get rid of square roots

$4x^2 - 8x + 4 = 12 - 4x$

$4x^2 - 4x - 8 = 0 \Longrightarrow x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

$x = 2$ or $x = -1$

Check your solution since there are multiple solutions:

Substitute $x = 2$ into the original equation:

$R.H.S = \sqrt{2+2} - \sqrt{3-2} = 2 - 1 = L.H.S \Longrightarrow x = 2$ is a solution

Substitute $x = -1$ into the original equation:

$R.H.S = \sqrt{2+(-1)} - \sqrt{3-(-1)} = 1 - 2 = -1 \neq L.H.S \Longrightarrow\ x = -1$ is NOT a solution

Example 4:

Without using a calculator, find the values of the **integers** a and b for which the solution of the equation $x\sqrt{24} = x\sqrt{2} + \sqrt{6}$ is $\frac{a+\sqrt{b}}{11}$.

[Why do we not substitute $x = \frac{a+\sqrt{b}}{11}$ into the equation? We will end up with an equation which we cannot apply "equality of surds"]

Solution:

Since $x = \frac{a+\sqrt{b}}{11}$ is a solution to the equation, we shall make x the subject first and then compare with the solution:

$$x\sqrt{24} = x\sqrt{2} + \sqrt{6} \Longrightarrow x = \frac{\sqrt{6}}{\sqrt{24}-\sqrt{2}}$$ ← Rationalize the denominator

$$x = \frac{\sqrt{6}}{\sqrt{24}-\sqrt{2}} \times \frac{\sqrt{24}+\sqrt{2}}{\sqrt{24}+\sqrt{2}} = \frac{\sqrt{6}\left(\sqrt{24}+\sqrt{2}\right)}{24-2}$$

$$x = \frac{\sqrt{6}\left(2\sqrt{6}+\sqrt{2}\right)}{22} = \frac{2(6)+\sqrt{6}\sqrt{2}}{22} = \frac{12+\sqrt{3}\sqrt{2}\sqrt{2}}{22} = \frac{12+2\sqrt{3}}{22} = \frac{6+\sqrt{3}}{11}$$

Compare $\frac{6+\sqrt{3}}{11}$ with $\frac{a+\sqrt{b}}{11}$ gives: $a = 6, b = 3$

Problems involving surds:

Example 1:

A cuboid has a square base of side $(\sqrt{3}+1)$ m and the height and volume of the cuboid respectively are $x\ m$ and $(x\sqrt{48}+\sqrt{12})\ m^3$ respectively . Find x, leaving your answer in the form of $a + b\sqrt{3}$, where a and b are integers.

Solution:

Volume of cuboid = (Area of square base)(Height of Cuboid)

$x\sqrt{48} + \sqrt{12} = x(\sqrt{3}+1)^2$ ← Find x

$x\sqrt{48} + \sqrt{12} = x(3 + 2\sqrt{3} + 1)$

$x\left(\sqrt{48}-2\sqrt{3}-4\right)=-\sqrt{12}$

$$x=-\frac{\sqrt{12}}{\sqrt{48}-2\sqrt{3}-4}=-\frac{\sqrt{12}}{4\sqrt{3}-2\sqrt{3}-4}=-\frac{\sqrt{12}}{2\sqrt{3}-4}$$

$$x=-\frac{\sqrt{12}}{2\sqrt{3}-4}\times\frac{2\sqrt{3}+4}{2\sqrt{3}+4}=-\frac{\sqrt{12}\left(2\sqrt{3}+4\right)}{\left(2\sqrt{3}-4\right)\left(2\sqrt{3}+4\right)}=-\frac{2\sqrt{3}\left(2\sqrt{3}+4\right)}{(4)(3)-16}=-\frac{12+8\sqrt{3}}{-4}$$

$$\therefore x=3+2\sqrt{3}$$

Example 2: (More Challenging)

If n is a positive integer, show that:

$$\frac{1}{\sqrt{n}+\sqrt{n+1}}=\sqrt{n+1}-\sqrt{n}$$

Hence, find the value of $\frac{1}{\sqrt{25}+\sqrt{26}}+\frac{1}{\sqrt{26}+\sqrt{27}}+\frac{1}{\sqrt{27}+\sqrt{28}}+\cdots+\frac{1}{\sqrt{99}+\sqrt{100}}$.

Solution:

$$L.H.S=\frac{1}{\sqrt{n}+\sqrt{n+1}}=\frac{\sqrt{n}-\sqrt{n+1}}{n-(n+1)}=-\left(\sqrt{n}-\sqrt{n+1}\right)=\sqrt{n+1}-\sqrt{n}=R.H.S$$

$$\underset{n=25}{\frac{1}{\sqrt{25}+\sqrt{26}}}+\underset{n=26}{\frac{1}{\sqrt{26}+\sqrt{27}}}+\underset{n=27}{\frac{1}{\sqrt{27}+\sqrt{28}}}+\cdots+\underset{n=99}{\frac{1}{\sqrt{99}+\sqrt{100}}}$$

$=\left(\sqrt{26}-\sqrt{25}\right)+\left(\sqrt{27}-\sqrt{26}\right)+\left(\sqrt{28}-\sqrt{27}\right)+\cdots+\left(\sqrt{100}-\sqrt{99}\right)$

$$\begin{aligned}&=\cancel{\sqrt{26}}-\sqrt{25}\\&+\sqrt{27}-\cancel{\sqrt{26}}\\&+\sqrt{28}-\cancel{\sqrt{27}}\\&\quad\vdots\\&+\sqrt{100}-\cancel{\sqrt{99}}\end{aligned}\Bigg\}=\sqrt{100}-\sqrt{25}$$

$\therefore$ Solution $=10-5=5$